



Epistemic dependence in AI-mediated learning

Yiran Du¹ · Yijia Yuan¹

Received: 19 June 2026 / Accepted: 22 July 2026
© The Author(s) 2026

Abstract

Artificial intelligence (AI) systems increasingly mediate how learners obtain explanations, synthesise sources, receive feedback and judge the quality of academic work. This critical-integrative review argues that the central educational question is not whether learners rely on AI, but whether that reliance preserves or displaces the epistemic work through which judgement develops. The review purposively bridges artificial intelligence in education, human–AI interaction, epistemic cognition, information behaviour, cognitive offloading, and social epistemology, while engaging a recent literature on epistemic authority, agency, infrastructure and over-reliance. It conceptualises epistemic dependence as reliance on AI for knowledge-related tasks and differentiates productive reliance from harmful dependence through six diagnostic criteria: contestability, recoverability, transfer, traceability, distributed responsibility and epistemic plurality. Four sociotechnical pathways are then analysed: fluent authority, frictionless delegation, opaque synthesis and institutionalised dependence. These pathways may affect verification, evaluative and disciplinary judgement, understanding, uncertainty tolerance, learner identity and epistemic justice, but the review does not assume that AI uniquely or uniformly causes these outcomes. It proposes relational epistemic agency as the normative aim: the capacity to question, verify, compare, justify and take responsibility for knowledge claims within human, technological and institutional relations. Design, pedagogical, assessment and governance implications are developed accordingly.

Keywords Artificial intelligence in education · Epistemic dependence · Epistemic agency · Generative AI · Human–AI interaction · Epistemic justice · Learner judgement

1 Introduction: from AI use to AI-mediated knowing

Artificial intelligence has moved from a specialist concern in artificial intelligence in education (AIED) to a general condition of learning. Intelligent tutoring systems, recommender systems, learning analytics dashboards, automated feedback tools, AI writing assistants and large language models now mediate routine educational activities. Reviews of AIED and AI in higher education document applications across assessment, prediction, personalisation, tutoring and student support (Chen et al. 2020a, b; Zawacki-Richter et al. 2019). Generative AI has intensified this trajectory, because one conversational interface can now combine brainstorming,

explanation, translation, coding, revision, feedback and research support (Chan and Hu 2023; Kasneci et al. 2023). The issue is therefore not simply whether AI is present in education, but what forms of knowing become normal when AI is embedded in searching, composing and evaluating.

Most educational discussion has been organised around adoption, effectiveness, academic integrity, policy, ethics or AI literacy. These frames remain necessary, but they can treat AI primarily as an intervention, productivity aid or compliance problem. Conversational AI also functions as an epistemic intermediary: it selects, compresses, reformulates and evaluates knowledge-like claims while presenting the result as a direct answer. In contrast with the conventional search, the learner may encounter an apparently integrated synthesis rather than a visible set of sources. The epistemic division of labour—the distribution of tasks involved in producing and warranting knowledge—becomes harder to inspect. Learners are consequently asked not only to use AI, but to decide when its output is credible, whose standards it

✉ Yiran Du
yd392@cam.ac.uk
Yijia Yuan
yy526@cam.ac.uk

¹ University of Cambridge, Cambridge, UK

embodies and where responsibility for accepting it should lie.

A fast-emerging literature now addresses this epistemic shift directly. Jose et al. (2025) describe generative AI as a possible surrogate knower and examine the displacement of educational authority. Chen (2025) frames generative AI as epistemic infrastructure and analyses whether systems sustain or supplant skilled epistemic action. Wu et al. (2025) propose a symbiotic learning partnership organised around adaptive epistemic stances, while Yan et al. (2025) synthesise emerging evidence on cognition, metacognition and epistemic agency. Zhai et al. (2024) review cognitive consequences associated with over-reliance on AI dialogue systems, and Rustambekov et al. (2025) discuss epistemic fragility in AI-integrated educational systems.

Pietarinen and Shi (2026) propose contestability, developmental integrity and value preservation as criteria for legitimate AI influence; Samuel (2026) theorises epistemic co-agency as a learner's reflexive regulation of AI influence; and Smith (2026) analyses formative epistemic injustice when AI use forecloses valuable developmental opportunities. The present review therefore does not claim to be the first to connect generative AI with epistemic agency or authority. Its contribution is more specific: it operationalises the boundary between productive reliance and harmful dependence, traces four mechanism-to-consequence pathways, and grounds the analysis in a relational account of agency that incorporates institutional responsibility and epistemic justice.

The central argument is that dependence is not itself an educational failure. Learning has always depended on teachers, peers, texts, instruments and institutions (Hardwig 1985). External support can extend cognition, widen access, reduce unnecessary load and create opportunities for reflection (Clark and Chalmers 1998; Clark 2025; Risko and Gilbert 2016). Dependence becomes problematic when the relation is difficult to contest, obscures its evidential basis, weakens the learner's capacity to reconstruct or transfer judgement, or allocates responsibility to people who lack meaningful control. Conversely, frequent AI use can remain productive when it provokes comparison, makes uncertainty visible and leaves the learner more capable of independent and collaborative judgement.

The review proceeds in five stages. It first states its critical-integrative method and positions the contribution against neighbouring work. It then conceptualises epistemic dependence and proposes diagnostic criteria for distinguishing productive from harmful forms. The central analysis examines four sociotechnical pathways linking AI affordances and institutional conditions to educational consequences. A normative section defends relational epistemic agency and addresses responsibility and justice. The final sections develop design, pedagogical, assessment, governance and research implications.

2 Review approach and scope

This article is a critical-integrative review rather than a systematic review or meta-analysis. Its purpose is conceptual synthesis and normative argument: to bring literatures that use different vocabularies into a common account of AI-mediated epistemic dependence. The corpus is purposive and illustrative rather than exhaustive. Searches were last updated on July 2026 in ERIC and Crossref metadata, supplemented by publisher-platform searches and reference-list inspection. Search combinations paired terms for generative AI, ChatGPT and AI dialogue systems with epistemic agency, epistemic authority, epistemic dependence, over-reliance, cognitive offloading, evaluative judgement, epistemic injustice and educational infrastructure. The review does not estimate aggregate effects or claim systematic coverage.

Titles and abstracts were screened for direct relevance to knowledge-related activity in teaching and learning; full texts were consulted when a source carried a substantive conceptual or evidential claim. Selection followed three criteria. First, priority was given to peer-reviewed studies and reviews that directly examined AI-mediated teaching or learning. Second, adjacent literatures were included when they supplied explanatory constructs not yet well developed in education-specific research, including trust in automation, digital credibility, cognitive offloading, explainability and social epistemology. Third, foundational philosophical and educational work was included where it clarified the normative meaning of dependence, agency, autonomy, judgement or justice. Technical studies concerned only with model performance, and educational studies concerned only with adoption or integrity, were not treated as central unless they illuminated an epistemic mechanism. Recent preprints and conference papers were used selectively and are identified as such. The final cited corpus comprises 60 works.

The synthesis was abductive and contrastive. Sources were read for claims about contextual conditions, mechanisms, forms of reliance, diagnostic markers, educational consequences and possible responses. Concepts were retained only when they performed distinct analytic work; overlapping mechanisms and risks were consolidated into four pathways. Close-neighbour analysis was used to specify the review's marginal contribution rather than to claim an empty field. Claims imported from non-educational domains are treated as explanatory hypotheses rather than direct evidence of educational effects. Evidence from Internet search, smartphones, medicine or aviation can render a mechanism intelligible without showing that the same effect occurs, at the same magnitude, in AI-mediated learning. The review's limitations are stated explicitly below.

3 Positioning the contribution against recent literature

The recent epistemic literature is convergent in recognising that generative AI is not a neutral tool. It can act as an authority, infrastructure, dialogue partner, cognitive shortcut or formative environment. The relevant novelty question is therefore not whether this review names an epistemic problem that no one else has noticed. It is whether it supplies distinctions and relations that permit the problem to be analysed more precisely. Table 1 groups the principal near-neighbours by their main contribution and clarifies the marginal contribution of the present review.

Three distinctions are especially important. First, instrumental assistance and epistemic dependence are not synonyms: the decisive issue is whether judgement is supported or delegated. Second, dependence is diagnosed through interactional and developmental markers rather than use frequency or a single error event. Third, the unit of analysis is sociotechnical. Learner dispositions matter, but so do assessment incentives, interface design, institutional procurement, commercial objectives and the distribution of authority among learners, teachers, institutions and technology providers. The review's novelty lies in combining these distinctions into an analytic test and pathway model, not in claiming that agency, authority or justice have been absent from the literature.

4 Conceptualising epistemic dependence

Epistemic dependence begins from the fact that much of what people know depends on others. Hardwig (1985) showed that modern knowledge is distributed across experts and institutions; Goldman (2001) asked how non-experts should decide which experts to trust. Digital environments extend authority beyond persons to platforms, rankings and infrastructures (Bartsch et al. 2025; Croce and Baghrmian 2024). In this sense, AI functions as an epistemic technology that reorganises knowledge practices rather than merely transmitting information (Alvarado 2023). AI-mediated learning gives this longstanding problem a distinctive form: learners rely on systems that can produce useful, fluent and personalised outputs but are not accountable knowers in the human sense and may conceal the sources, values and uncertainty underlying their responses.

For this review, epistemic dependence refers to reliance on AI systems for knowledge-related activity, including obtaining information, receiving explanations, synthesising sources, evaluating quality, selecting interpretations,

Table 1 Positioning the review against close recent contributions

Contribution	Primary focus	Differentiation of the present review
Authority and infrastructure: Jose et al. (2025), Chen (2025)	AI as surrogate knower or epistemic infrastructure; displacement of educational authority and reorganisation of skilled epistemic action	Operationalises when authority-support becomes judgement-displacing dependence and links infrastructure to cross-level responsibility
Agency and co-agency: Wu et al. (2025), Yan et al. (2025), Samuel (2026)	Adaptive epistemic stances, empirical patterns of epistemic agency, and reflexive human-AI co-agency or epistemic sovereignty	Specifies six diagnostic criteria and a developmental test for whether relational participation is preserved across tasks and institutions
Over-reliance and fragility: Zhai et al. (2024), Rustambekov et al. (2025)	Cognitive consequences of over-reliance and epistemic fragility in AI-integrated educational systems	Distinguishes use frequency from epistemic dependence, balances risk with productive reliance, and treats transferred evidence cautiously
Normative thresholds: Pietarinen and Shi (2026)	Contestability, developmental integrity and value preservation as constraints on manipulation and deception in AI-mediated education	Connects normative thresholds to mechanism-to-consequence pathways, recoverability, transfer and distributed institutional responsibility
Formative epistemic injustice: Smith (2026)	Educational wrongs arising when GenAI use restricts knowledge, practice or learners' accurate self-assessment	Integrates formative harm with plurality, learner identity and the allocation of responsibility across providers, institutions and educators
Present review	Critical integrative review of AI-mediated learning as an epistemic relation	Contributes the productive-reliance/harmful-dependence boundary, four pathways, and a relational account of epistemic agency, justice and responsibility

resolving uncertainty and justifying claims. This definition is broader than behavioural frequency. A learner may use AI repeatedly yet remain epistemically agentic if the interaction prompts questioning, comparison and revision. Another learner may use AI once but treat its answer as decisive in a high-stakes evaluative task. The analytic boundary is therefore not use versus non-use. It is the relation between assistance and judgement.

Two forms of assistance are especially useful analytically. Instrumental or representational assistance helps the learner generate options, translate, format, retrieve, summarise or re-express material. Judgement-bearing assistance evaluates correctness, relevance, quality, persuasiveness, ethical acceptability or evidential sufficiency. The categories can overlap: a summary already selects what matters, and brainstorming can shape the space of plausible ideas. Nevertheless, judgement-bearing assistance is normatively more demanding, because it delegates not only production but the standards by which production is assessed. The distinction is therefore the paper's central analytic pivot.

Epistemic trust and epistemic authority clarify why the distinction matters. Trust concerns a disposition to rely on a source; authority concerns a source's perceived competence and legitimacy in settling a question. Conversational systems can invite trust through responsiveness, social presence and confidence rather than through accountable expertise (Sahebi and Formosa 2025; Sedlakova et al. 2025). They can also influence belief revision by shaping what users regard as plausible, doubtful or worth investigating (Coeckelbergh 2026). The educational issue is not that learners trust at all, but whether the relation permits trust to be calibrated, challenged and revised.

Epistemic cognition research adds a developmental perspective. Learners' beliefs and practices concerning knowledge, evidence, certainty and justification shape how they reason and learn (Hofer and Pintrich 1997; Chinn et al. 2011;

Greene and Yu 2016). These practices are associated with achievement and can be changed through instruction (Cartiff et al. 2021; Greene et al. 2018). They also involve emotions: uncertainty, curiosity, confusion and frustration can either support inquiry or prompt premature closure (Muis et al. 2018). AI is therefore not merely a source environment. It becomes part of the context in which epistemic dispositions and habits develop.

The counterfactual must be specified carefully. Learners often struggled with source evaluation, verification and uncertainty before generative AI (Hargittai et al. 2010; McGrew et al. 2018; Metzger 2007). AI should not be blamed for creating a deficit that predates it. A claim of harm requires comparison with a relevant baseline: the learner's prior practice, a plausible non-AI workflow, or the developmental trajectory expected under instruction. AI may improve epistemic performance for learners who otherwise lack accessible explanations, language support or feedback. It becomes concerning when it makes existing weaknesses less visible, reduces practice of a target capability, or shifts a learner away from a better available trajectory. Effects will therefore vary by learner, task, discipline and design.

Table 2 operationalises the productive-reliance/harmful-dependence boundary. The criteria synthesise work on appropriate reliance and calibration (Parasuraman and Riley 1997; Hoff and Bashir 2015; Lee and See 2004), reasoning and contestability (Baum et al. 2022; Pietarinen and Shi 2026), relational agency (Wu et al. 2025; Samuel 2026) and institutional epistemic justice (Anderson 2012; Fricker 2007; Kay et al. 2024; Smith 2026). They are not a psychometric scale and need empirical validation. They function as diagnostic questions for research, design and pedagogy. No single criterion is decisive in every context; the pattern matters, as does the importance of the delegated judgement.

These criteria also resolve the apparent tension with extended cognition. The extended-mind thesis shows that

Table 2 Diagnostic criteria for productive epistemic reliance and harmful epistemic dependence

Criterion	Productive epistemic reliance	Harmful epistemic dependence
Contestability	The learner can question, reject or revise the output; alternatives are visible and disagreement is pedagogically supported	The output functions as a default or final authority; interface, policy or assessment discourages challenge
Recoverability	The learner can reconstruct the main reasoning, evidence or procedure with reduced support	The learner can reproduce a product but cannot explain how the conclusion was reached or warranted
Transfer	Capability persists or improves when the system is absent, changed or applied to a new problem	Performance collapses when AI support is withdrawn or when a superficially different task is encountered
Traceability	Claims can be connected to inspectable sources, assumptions, transformations and uncertainty	The synthesis hides its evidential structure, leaving the learner unable to audit provenance or limitations
Distributed responsibility	Learners, teachers, institutions and providers have responsibilities proportionate to their control and knowledge	Learners bear accountability for outputs while institutions or providers control opaque systems and incentives
Epistemic plurality	The interaction exposes relevant disagreement, disciplinary standards and marginalised perspectives	The system naturalises a dominant answer, language or knowledge hierarchy as neutral and comprehensive

cognitive achievement can be distributed across people and artefacts (Clark and Chalmers 1998), and generative AI may become a powerful component of such systems (Clark 2025). Offloading is not the problem as such. It extends agency when the learner can exercise control over the coupling, interrogate what the system contributes, recover the relevant reasoning at the level required by the educational aim and transfer capability across contexts. It bypasses agency when the system performs judgement in a way that is opaque, difficult to contest and developmentally substitutive. Samuel's (2026) notion of epistemic co-agency is compatible with this asymmetry: AI shapes the reasoning environment, but the learner must retain responsibility for evaluating and warranting claims. The question is not whether cognition is internal, but whether the human-technology relation preserves meaningful epistemic participation.

5 From affordances to consequences: four pathways of epistemic dependence

This review identifies four parallel pathways linking contextual conditions to configurations of epistemic reliance. A pathway is not a deterministic causal chain. It specifies a plausible relation to be investigated: an affordance or institutional condition changes how epistemic work is distributed; the resulting relation is assessed through six diagnostic criteria; and the configuration is interpreted as productive reliance or harmful dependence. Educational consequences are attached only to harmful configurations, while design, pedagogical and governance responses may intervene at more than one point in the model. The framework also distinguishes evidence levels. Direct educational studies warrant claims about observed AI use; adjacent research from search, smartphones, automation or medicine provides hypotheses and design warnings, not proof of equivalent educational effects.

Epistemic friction refers here to the effort and resistance involved in finding, comparing, interpreting and justifying claims. Some friction is wasteful, exclusionary or unrelated to the learning goal; removing it can increase access and free capacity for higher-level reasoning. Other friction is productive, because it creates occasions to notice uncertainty, encounter alternatives and practise judgement. The design question is therefore not whether AI makes learning easier, but which forms of effort it removes and what epistemic work replaces them.

Contextual conditions shape four parallel pathways rather than a sequence among them. The diagnostic criteria then distinguish productive epistemic reliance from harmful epistemic dependence. Only harmful configurations are connected to the specified educational consequences. Design, pedagogical and governance responses can modify

contextual conditions, pathways and configurations, while relational epistemic agency provides the normative orientation of the model rather than an automatic endpoint.

6 Pathway 1: fluent authority and personalised social presence

Large language models produce grammatically polished, coherent and often confident responses. Fluency is epistemically persuasive, because users can mistake ease of processing for reliability. The educational risk is not error alone but the alignment of error, confidence and immediacy. Reviews of generative AI in education document both explanatory usefulness and vulnerability to inaccurate or fabricated content (Farrokhnia et al. 2024; Kasneci et al. 2023; Rahman and Watanobe 2023). Direct evidence about how particular learners calibrate trust remains uneven, so the stronger claim is that fluent presentation gives reason to expect miscalibration, especially when learners have limited domain knowledge.

Anthropomorphism, social presence and personalisation can amplify this effect. Conversational systems occupy roles associated with human epistemic others: respondent, tutor, editor, adviser and evaluator. A supportive tone, apparent memory and adaptation to the learner's level may increase engagement and reduce anxiety. These are genuine educational benefits. Yet, the same features can encourage transfer of interpersonal trust to a system that lacks human understanding, responsibility and commitment to the learner (Sahebi and Formosa 2025; Sedlakova et al. 2025). Personalisation can also narrow exposure to resistant sources or disciplinary standards that are not tailored to immediate preference.

The principal consequences are misplaced epistemic authority and normative displacement. Learners may ask AI not only what is true, but what counts as a strong argument, an appropriate interpretation, an ethical response or acceptable academic quality. When those standards are absorbed without disciplinary mediation, the system shapes the norms by which knowledge work is evaluated. Productive reliance remains possible when dialogue is designed for contestation: the system presents alternatives, marks uncertainty, invites reasons for disagreement and directs the learner back to accountable sources and teachers.

7 Pathway 2: frictionless delegation and reduced epistemic work

Generative AI compresses several activities—searching, reading, comparing, synthesising and drafting—into a single request–response cycle. Cognitive offloading can

be adaptive when it frees limited resources for planning, reflection or complex problem solving (Risko and Gilbert 2016). However, adjacent research suggests that persistent access to external information can alter memory, confidence and future search behaviour (Sparrow et al. 2011; Storm et al. 2017; Fisher et al. 2015). These studies do not establish equivalent effects for generative AI in education. They render intelligible a hypothesis: when intermediate epistemic actions disappear from the workflow, learners may complete tasks without practising the actions that the task was intended to develop.

Assessment incentives intensify the pathway. If institutions reward polished products while leaving process and justification invisible, delegating epistemic work is rational. AI can then shift from scaffold to substitute, especially in feedback and evaluation. A learner who asks whether an argument is strong or a source is relevant delegates evaluative judgement itself. Educational research treats evaluative judgement as a capability to make decisions about the quality of one's own and others' work (Tai et al. 2018), while feedback literacy includes making sense of information and acting on it rather than merely receiving corrections (Carless and Boud 2018). Automated feedback supports learning when it becomes material for judgement; it displaces learning when its verdict replaces that judgement.

Possible consequences include shallow understanding, reduced tolerance of uncertainty and weak transfer. Zhai et al. (2024) report concerns about over-reliance and cognitive abilities, but their review also reflects a young and heterogeneous evidence base. The more precise research question is whether learners can explain the reasoning, perform the task under reduced support and adapt their knowledge to a new problem. Productive friction can be reintroduced through delayed answers, prediction before assistance, comparison of multiple outputs, source checking and reflective justification.

8 Pathway 3: opaque synthesis and hidden evidential structure

Conversational AI often presents a synthesis without making its source selection, weighting, exclusions or uncertainty inspectable. Opacity is therefore not only a property of model internals. It is an educational condition in which the learner cannot see how an answer relates to evidence. Research on explainable AI and epistemic opacity shows why reason-giving and provenance matter when users must remain responsible for decisions (Baum et al. 2022; Hähnel and Hauswald 2025; Schmidt et al. 2025). Medical and other high-stakes studies cannot be transferred directly to education, but they highlight a responsibility problem that

is especially salient, while learners are developing practices of justification.

Traditional search was not transparent or neutral: rankings, interface cues and commercial objectives already shaped attention. The counterfactual difference is narrower. Search commonly exposes multiple documents, authors and domains, even if learners evaluate them poorly. Conversational synthesis can hide that plurality behind one voice. The learner's task changes from selecting among visible sources to recovering the evidential structure of an answer that appears already integrated. This can weaken verification while improving the surface quality of the product, making epistemic deficits harder for teachers to detect.

The pathway also threatens disciplinary reasoning. Disciplines differ in what counts as evidence, explanation and warranted inference (Goldman et al. 2016). A generic synthesis may flatten these standards: historical claims require provenance and contextualisation; scientific claims require methods, measurement and causal reasoning; philosophical claims require conceptual precision and argumentative validity. Productive reliance requires traceability at the level relevant to the discipline, not a promise of complete model transparency. Systems should connect claims to inspectable sources, expose disagreement and make the learner articulate why a source or inference is acceptable.

9 Pathway 4: institutionalised dependence and distributed authority

Epistemic dependence is not produced only by individual preference. Universities and schools shape it through procurement, platform integration, assessment design, timetabling, staffing, policy and professional development. Commercial systems also encode objectives concerning engagement, speed, cost and data capture. Once AI is built into learning management systems, writing environments, library interfaces or feedback workflows, it may become the default route into academic work rather than a discrete tool chosen by the learner. Chen's (2025) infrastructure perspective is important here: systems reorganise the conditions under which epistemic action is possible, not merely the efficiency of isolated tasks.

Institutionalisation redistributes authority and responsibility. A student may be held accountable for a claim generated through an institutionally licenced but opaque system; a teacher may be expected to police use without access to system logs or procurement decisions; a university may depend on vendor assurances that cannot be independently audited. An adequate account must therefore avoid assigning all responsibility to the end user. Responsibility should track control, knowledge and benefit across learners, educators, institutions and providers.

This pathway also makes epistemic justice central rather than peripheral. Generative systems draw on unequal knowledge infrastructures and may reproduce dominant languages, classifications and perspectives while marginalising local, minoritised or experiential knowledge (Kay et al. 2024). Epistemic injustice concerns not only access to information but whose testimony is credible, which interpretive resources are available and which developmental opportunities remain open (Fricker 2007; Räisä and Stocchetti 2024; Smith 2026). Institutional dependence can therefore amplify hidden hierarchies under the appearance of a neutral answer. It can also affect learner identity: repeated reliance may foster a self-conception of being unable to write, understand or judge without AI. Productive reliance should expand participation and capability over time; harmful dependence makes learners and institutions more fragile when the system is absent, changes its terms or fails particular communities.

10 Normative core: relational epistemic agency, responsibility and justice

This review positions relational epistemic agency as the normative orientation of AI-mediated learning, but this aim requires a substantive defence. It is not an automatic consequence of using AI or of applying a discrete design intervention. The value at stake is not independence from others or tools. Education necessarily involves dependence, and ideals of self-sufficient knowing are both unrealistic and exclusionary. The relevant conception is relational and scaffolded: learners exercise agency through social, material and institutional supports while retaining meaningful capacities to participate in how claims are formed, assessed and accepted. This position extends work on relational autonomy and hybrid human–AI learning, which treats agency as socially enabled and technologically mediated rather than as isolation from dependence (Mackenzie and Stoljar 2000; Markauskaite et al. 2022; Molenaar 2022; Gašević et al. 2023).

Epistemic agency matters, because education is not only the production of correct outputs. It is the formation of people who can participate responsibly in knowledge practices: asking questions, recognising uncertainty, assessing evidence, revising beliefs, giving reasons and responding to challenge. These capabilities are constitutive of disciplinary participation, democratic judgement and lifelong learning. A system can improve immediate performance while undermining this developmental aim if it removes the learner from the practices by which standards are understood and claims become answerable.

Relational agency supplies a clear answer to the extended-mind tension. External resources extend agency when they are integrated under conditions of contestability,

recoverability, transfer and appropriate control. A calculator can support mathematical agency when the learner understands when and why its operations are appropriate; a reference manager can support scholarship when source selection remains accountable; a generative model can support inquiry when it expands hypotheses, reveals alternatives and prompts verification. The same tools bypass agency when they become non-contestable authorities or when the learner cannot recover the reasoning required by the educational objective. Pietarinen and Shi's (2026) concern with developmental integrity sharpens the test: legitimate support should preserve the formation of reflective judgement rather than merely secure immediate performance. The normative test is functional and developmental, not biological: does the relation enlarge the learner's capacity to participate in epistemic practice, or merely deliver a product?

Responsibility is the second component. Learners should take responsibility for claims they submit, but responsibility is fair only when they have sufficient knowledge and control. Institutions that mandate or normalise AI, teachers who design AI-dependent assessments, and providers that control model behaviour also shape the epistemic relation. A relational account therefore replaces the fiction of the isolated user with distributed responsibility. Governance should specify who is answerable for provenance, errors, accessibility, data use, model updates and the conditions under which AI-mediated work is assessed.

Justice is the third component. Fricker's (2007) account of epistemic injustice identifies harms to people in their capacity as knowers, while Anderson (2012) emphasises epistemic justice as a virtue of social institutions. Smith's (2026) account of formative epistemic injustice adds a specifically educational concern: learning arrangements can wrong students by restricting the knowledge, practice and accurate self-assessment through which they develop as knowers. These perspectives show why agency cannot be reduced to individual critical thinking. Learners differ in language, disability, prior educational opportunity, cultural knowledge and access to human support. AI can widen participation by translating, explaining and reducing barriers, but it can also discount accents, local knowledge or non-dominant forms of expression. Epistemically responsible design must therefore preserve plurality, make contestation possible and ensure that marginalised learners are not required to accept the system's categories in order to participate.

The resulting norm is relational epistemic agency: the capacity to question, verify, compare, justify and know responsibly within relations of interdependence. It overlaps with accounts of shared or epistemic co-agency (Wu et al. 2025; Samuel 2026), but retains an explicit asymmetry: AI systems may shape and extend reasoning without possessing reciprocal responsibility or legitimate authority. The norm is compatible with substantial AI use. It rejects both

technological solutionism and a simple anti-dependence position. The practical objective is to configure dependence, so that learners gain epistemic capabilities, institutions remain accountable and knowledge plurality is not silently narrowed.

11 Implications for design, pedagogy, assessment and governance

Responses should not be confined to remediating harm after it occurs. Design can alter the contextual conditions and pathways through which dependence is produced by supporting inspectable and contestable use rather than frictionless acceptance. First, systems should provide claim-level provenance where feasible, distinguish retrieved evidence from model-generated synthesis, and make source limitations visible. Second, uncertainty should be represented through alternatives, ranges, explicit unknowns, and prompts to seek human or disciplinary expertise, rather than through an unsupported confidence score. Third, interfaces should preserve user control: learners need to revise prompts, compare outputs, reject recommendations, and understand when personalisation is shaping the response. Fourth, systems should incorporate epistemic plurality by surfacing disagreement, culturally and linguistically diverse sources, and the boundaries of the training data. These requirements do not demand complete model transparency; they demand sufficient traceability for educational judgement.

Pedagogy should move from policing AI use to teaching AI-mediated judgement. Students need routines for lateral reading, source triangulation, claim verification and comparison of competing explanations (McGrew 2020; Wineburg and McGrew 2019). Assignments can require learners to annotate an AI response, identify unsupported assumptions, recover sources, explain what evidence would change their view and document why they accepted or rejected particular suggestions. Teachers can use adversarial questions: What is missing? Whose perspective is absent? Which disciplinary standard is being applied? What would make this conclusion false? Such activities connect AI literacy to epistemic practice rather than reducing it to prompt technique (Ng et al. 2021), and they treat AI output as an object of inquiry rather than an answer key.

Feedback practices require particular care. AI-generated comments can increase access to timely feedback and reduce barriers to revision. However, feedback becomes educationally valuable only when learners interpret, evaluate and act on it. Tasks should therefore require comparison between AI, peer, teacher and exemplar-based feedback, followed by a justified revision decision. This aligns AI use with feedback literacy and evaluative judgement (Carless and Boud 2018; Tai et al. 2018). The learner should not merely implement

corrections but explain which advice was warranted, which was not and how the decision relates to disciplinary criteria.

Assessment should make process, judgement and transfer visible. If only polished products are rewarded, dependence is predictable. Assessment can include staged drafts, oral defence, source maps, reflective decision logs, in-class verification, comparison tasks and occasions to perform a core capability with reduced AI support. Such designs should not become surveillance-heavy or disadvantage students who use assistive technologies. The purpose is to sample whether learning outcomes remain recoverable and transferable, not to enforce an unrealistic condition of technological purity.

Teachers should be recognised as epistemic mediators rather than misconduct monitors. Their role is to make standards visible: what counts as evidence in a discipline, how uncertainty should be represented, when AI is an appropriate source of assistance, and what forms of disclosure are required. Professional development should therefore be discipline-specific and include opportunities to inspect system limitations, design contestable tasks and discuss responsibility. The teacher as epistemic mentor is not displaced by AI; the role becomes more important when automated systems present authoritative-seeming judgements (Jose et al. 2025). This emphasis is consistent with human–AI learning frameworks that place educator mediation and learner capability development at the centre (Markauskaite et al. 2022; Molenaar 2022; Gašević et al. 2023).

Institutional governance should address the production of dependence. Procurement processes should evaluate provenance, accessibility, bias, data governance, model-update practices, auditability and the vendor's capacity to support educational objectives. Policies should distinguish legitimate reliance from prohibited substitution, specify which learning outcomes must remain demonstrably human, and allocate responsibility across stakeholders (Chan 2023). Institutions should also monitor whether AI integration changes staffing, feedback expectations or assessment in ways that make use effectively compulsory. Responsible AI policy cannot place all verification burdens on individual learners while leaving infrastructural decisions unexamined.

12 Future research

The empirical agenda should ask which epistemic actions are preserved, transformed or displaced, rather than whether AI helps or harms learning in the aggregate. Process evidence is essential. Screen recordings, interaction traces, think-aloud protocols, interviews and classroom observations can show whether learners compare sources, revise beliefs, recognise uncertainty and contest AI recommendations. Product quality alone cannot reveal whether judgement was exercised or bypassed.

The diagnostic criteria proposed here require operationalisation and validation. Research should develop measures of contestability, recoverability, transfer, traceability, distributed responsibility and epistemic plurality. Experimental and quasi-experimental studies can compare AI-supported workflows with relevant non-AI baselines, rather than assuming that pre-AI practice was epistemically strong. Withdrawal and transfer designs are particularly important: can learners explain and perform the relevant reasoning when support is reduced, and do they improve over time?

Direct educational testing is needed for claims currently inferred from adjacent domains. Studies should examine whether fluent AI output produces the same trust effects as human or conventional digital sources; whether compressed search and synthesis alter memory or verification; and how opacity affects responsibility in different disciplines. Longitudinal research should investigate habit formation, learner identity and institutional lock-in. Effects should be disaggregated by prior knowledge, language, disability, socioeconomic context and discipline, because the same AI affordance may scaffold one learner while displacing another's judgement.

Finally, research should examine organisations and markets, not only learners. Procurement decisions, platform defaults, assessment regimes and vendor business models may determine dependence more powerfully than individual attitudes. Comparative institutional studies can analyse how authority and responsibility are distributed, whose knowledge is represented, and whether governance arrangements permit meaningful contestation. The field needs evidence about the kinds of educational systems—not only the kinds of users—that sustain relational epistemic agency.

13 Limitations

This review has four limitations. First, it is a purposive critical review and does not claim exhaustive coverage or permit prevalence estimates; the documented search procedure improves auditability but is not equivalent to systematic screening. Second, the field is moving rapidly. The search was updated to July 2026, yet close contributions may appear after that cut-off, and several cited works are preprints, editorials or conference papers whose status and arguments may change. Third, direct evidence on long-term epistemic development remains limited, so some pathways are theoretically grounded hypotheses rather than established causal effects. Fourth, the framework abstracts across systems and disciplines. Generative models differ in retrieval, provenance, personalisation and governance, while educational tasks differ in what must be learned and what may legitimately be offloaded. The framework should therefore

be used to formulate context-sensitive questions, not as a universal prediction that AI use produces dependence.

14 Conclusion

AI-mediated learning is an epistemic transformation, but not a uniform decline in human knowing. Learners have always depended on people, institutions and artefacts, and generative AI can extend access, explanation and reflection. The educational problem is the configuration of that dependence. Fluent authority, frictionless delegation, opaque synthesis and institutionalised dependence can displace verification and judgement when outputs are difficult to contest, reasoning cannot be recovered, capability does not transfer, and responsibility is unfairly allocated. The same pathways can support productive reliance when systems expose evidence, preserve plurality, invite challenge and leave learners more capable over time. Design, pedagogical and governance responses must therefore reshape the conditions and relations through which dependence develops, not merely correct downstream errors. The normative task is to cultivate relational epistemic agency: not independence from the machine, but the capacity to judge responsibly with, through and against the systems that mediate knowledge.

Funding No funding was received for this work.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Ethics approval Not applicable.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Alvarado R (2023) AI as an epistemic technology. *Sci Eng Ethics* 29:Article 32. <https://doi.org/10.1007/s11948-023-00451-3>

- Anderson E (2012) Epistemic justice as a virtue of social institutions. *Soc Epistemol* 26(2):163–173. <https://doi.org/10.1080/02691728.2011.652211>
- Bartsch A, Neuberger C, Stark B, Karnowski V, Maurer M, Pentzold C, Quandt T, Quiring O, Schemer C (2025) Epistemic authority in the digital public sphere: an integrative conceptual framework and research agenda. *Commun Theory* 35(1):37–50. <https://doi.org/10.1093/ct/qtae020>
- Baum K, Mantel S, Schmidt E, Speith T (2022) From responsibility to reason-giving explainable artificial intelligence. *Philos Technol* 35:Article 12. <https://doi.org/10.1007/s13347-022-00510-w>
- Carless D, Boud D (2018) The development of student feedback literacy: enabling uptake of feedback. *Assess Eval High Educ* 43(8):1315–1325. <https://doi.org/10.1080/02602938.2018.1463354>
- Cartiff BM, Duke RF, Greene JA (2021) The effect of epistemic cognition interventions on academic achievement: a meta-analysis. *J Educ Psychol* 113(3):477–498. <https://doi.org/10.1037/edu0000490>
- Chan CKY (2023) A comprehensive AI policy education framework for university teaching and learning. *Int J Educ Technol High Educ* 20:Article 38. <https://doi.org/10.1186/s41239-023-00408-3>
- Chan CKY, Hu W (2023) Students' voices on generative AI: perceptions, benefits, and challenges in higher education. *Int J Educ Technol High Educ* 20:Article 43. <https://doi.org/10.1186/s41239-023-00411-8>
- Chen B (2025) Beyond tools: generative AI as epistemic infrastructure in education [Preprint]. <https://doi.org/10.48550/arXiv.2504.06928>
- Chen L, Chen P, Lin Z (2020a) Artificial intelligence in education: a review. *IEEE Access* 8:75264–75278. <https://doi.org/10.1109/ACCESS.2020.2988510>
- Chen X, Xie H, Zou D, Hwang G-J (2020b) Application and theory gaps during the rise of artificial intelligence in education. *Comput Educ Artif Intell* 1:Article 100002. <https://doi.org/10.1016/j.caeai.2020.100002>
- Chinn CA, Buckland LA, Samarapungavan A (2011) Expanding the dimensions of epistemic cognition: arguments from philosophy and psychology. *Educ Psychol* 46(3):141–167. <https://doi.org/10.1080/00461520.2011.587722>
- Clark A (2025) Extending minds with generative AI. *Nat Commun* 16:Article 4627. <https://doi.org/10.1038/s41467-025-59906-9>
- Clark A, Chalmers D (1998) The extended mind. *Analysis* 58(1):7–19. <https://doi.org/10.1093/analys/58.1.7>
- Coeckelbergh M (2026) AI and epistemic agency: how AI influences belief revision and its normative implications. *Soc Epistemol* 40(1):59–71. <https://doi.org/10.1080/02691728.2025.2466164>
- Croce M, Baghrmian M (2024) Experts-Part II: the sources of epistemic authority. *Philos Compass* 19(9–10):Article e70005. <https://doi.org/10.1111/phc3.70005>
- Farrokhnia M, Banihashem SK, Noroozi O, Wals A (2024) A SWOT analysis of ChatGPT: implications for educational practice and research. *Innov Educ Teach Int* 61(3):460–474. <https://doi.org/10.1080/14703297.2023.2195846>
- Fisher M, Goddu MK, Keil FC (2015) Searching for explanations: how the Internet inflates estimates of internal knowledge. *J Exp Psychol Gen* 144(3):674–687. <https://doi.org/10.1037/xge0000070>
- Fricker M (2007) *Epistemic injustice: power and the ethics of knowing*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780198237907.001.0001>
- Gašević D, Siemens G, Sadiq S (2023) Empowering learners for the age of artificial intelligence. *Comput Educ Artif Intell* 4:Article 100130. <https://doi.org/10.1016/j.caeai.2023.100130>
- Goldman AI (2001) Experts: which ones should you trust? *Philos Phenomenol Res* 63(1):85–110. <https://doi.org/10.1111/j.1933-1592.2001.tb00093.x>
- Goldman SR, Britt MA, Brown W, Cribb G, George M, Greenleaf C, Lee CD, Shanahan C, Project READI (2016) Disciplinary literacies and learning to read for understanding: a conceptual framework for disciplinary literacy. *Educ Psychol* 51(2):219–246. <https://doi.org/10.1080/00461520.2016.1168741>
- Greene JA, Yu SB (2016) Educating critical thinkers: the role of epistemic cognition. *Policy Insights Behav Brain Sci* 3(1):45–53. <https://doi.org/10.1177/2372732215622223>
- Greene JA, Duke RF, Cartiff BM (2018) A meta-analytic review of the relationship between epistemic cognition and academic achievement. *J Educ Psychol* 110(8):1084–1111. <https://doi.org/10.1037/edu0000263>
- Hähnel M, Hauswald R (2025) Trust and opacity in artificial intelligence: mapping the discourse. *Philos Technol* 38:Article 115. <https://doi.org/10.1007/s13347-025-00947-9>
- Hardwig J (1985) Epistemic dependence. *J Philos* 82(7):335–349. <https://doi.org/10.2307/2026523>
- Hargittai E, Fullerton L, Menchen-Trevino E, Thomas KY (2010) Trust online: young adults' evaluation of web content. *Int J Commun* 4:468–494
- Hofer BK, Pintrich PR (1997) The development of epistemological theories: beliefs about knowledge and knowing and their relation to learning. *Rev Educ Res* 67(1):88–140. <https://doi.org/10.3102/00346543067001088>
- Hoff KA, Bashir M (2015) Trust in automation: integrating empirical evidence on factors that influence trust. *Hum Factors* 57(3):407–434. <https://doi.org/10.1177/0018720814547570>
- Jose B, Cleetus A, Joseph B, Joseph L, Jose B, John AK (2025) Epistemic authority and generative AI in learning spaces: rethinking knowledge in the algorithmic age. *Front Educ* 10:1647687. <https://doi.org/10.3389/feduc.2025.1647687>
- Kasneci E, Sessler K, Küchemann S, Bannert M, Dementieva D, Fischer F, Gasser U, Groh G, Günnemann S, Hüllermeier E, Krusche S, Kutyniok G, Michaeli T, Nerdel C, Pfeffer J, Poquet O, Sailer M, Schmidt A, Seidel T, Stadler M, Weller J, Kuhn J, Kasneci G (2023) ChatGPT for good? On opportunities and challenges of large language models for education. *Learn Individ Differ* 103:102274. <https://doi.org/10.1016/j.lindif.2023.102274>
- Kay J, Kasirzadeh A, Mohamed S (2024) Epistemic injustice in generative AI. *Proc AAAI/ACM Conf AI Ethics Soc* 7(1):684–697. <https://doi.org/10.1609/aies.v7i1.31671>
- Lee JD, See KA (2004) Trust in automation: designing for appropriate reliance. *Hum Factors* 46(1):50–80. <https://doi.org/10.1518/hfes.46.1.50.30392>
- Mackenzie C, Stoljar N (eds) (2000) *Relational autonomy: feminist perspectives on autonomy, agency, and the social self*. Oxford University Press. <https://doi.org/10.1093/oso/9780195123333.001.0001>
- Markauskaite L, Marrone R, Poquet O, Knight S, Martinez-Maldonado R, Howard S, Tondeur J, De Laat M, Buckingham Shum S, Gašević D, Siemens G (2022) Rethinking the entwinement between artificial intelligence and human learning: what capabilities do learners need for a world with AI? *Comput Educ Artif Intell* 3:100056. <https://doi.org/10.1016/j.caeai.2022.100056>
- McGrew S (2020) Learning to evaluate: an intervention in civic online reasoning. *Comput Educ* 145:103711. <https://doi.org/10.1016/j.compedu.2019.103711>
- McGrew S, Breakstone J, Ortega T, Smith M, Wineburg S (2018) Can students evaluate online sources? Learning from assessments of civic online reasoning. *Theory Res Soc Educ* 46(2):165–193. <https://doi.org/10.1080/00933104.2017.1416320>
- Metzger MJ (2007) Making sense of credibility on the web: models for evaluating online information and recommendations for future research. *J Am Soc Inf Sci Technol* 58(13):2078–2091. <https://doi.org/10.1002/asi.20672>

- Molenaar I (2022) Towards hybrid human–AI learning technologies. *Eur J Educ* 57(4):632–645. <https://doi.org/10.1111/ejed.12527>
- Muis KR, Chevrier M, Singh CA (2018) The role of epistemic emotions in personal epistemology and self-regulated learning. *Educ Psychol* 53(3):165–184. <https://doi.org/10.1080/00461520.2017.1421465>
- Ng DTK, Leung JKL, Chu SKW, Qiao MS (2021) Conceptualising AI literacy: an exploratory review. *Comput Educ Artif Intell* 2:Article 100041. <https://doi.org/10.1016/j.caeai.2021.100041>
- Parasuraman R, Riley V (1997) Humans and automation: use, misuse, disuse, abuse. *Hum Factors* 39(2):230–253. <https://doi.org/10.1518/001872097778543886>
- Pietarinen A-V, Shi S (2026) Manipulation and deception in generative AI-mediated education: preserving epistemic agency, critical thinking, and creativity. *Postdigit Sci Educ*. <https://doi.org/10.1007/s42438-026-00644-6>. (advance online publication)
- Rahman MM, Watanobe Y (2023) Chatgpt for education and research: opportunities, threats, and strategies. *Appl Sci* 13(9):5783. <https://doi.org/10.3390/app13095783>
- Räisä T, Stocchetti M (2024) Epistemic injustice and education in the digital age: introduction. *J Digit Soc Res* 6(3):1–9. <https://doi.org/10.33621/jdsr.v6i3.33235>
- Risko EF, Gilbert SJ (2016) Cognitive offloading. *Trends Cogn Sci* 20(9):676–688. <https://doi.org/10.1016/j.tics.2016.07.002>
- Rustambekov I, Bazarov S, Mamanazarov S, Juraev J, Panabergenova J, Musaev S (2025) The phenomenon of “epistemic fragility” in educational systems with integrated AI: Investigating the growing dependence on external intelligent systems. In: 2025 5th international conference on technology enhanced learning in higher education (TELE). IEEE, pp 42–47. <https://doi.org/10.1109/TELE66816.2025.11212068>
- Sahebi S, Formosa P (2025) The AI-mediated communication dilemma: epistemic trust, social media, and the challenge of generative artificial intelligence. *Synthese* 205:128. <https://doi.org/10.1007/s11229-025-04963-2>
- Samuel A (2026) Learning with machines: toward a theory of epistemic co-agency. *Comput Educ Artif Intell* 10:100573. <https://doi.org/10.1016/j.caeai.2026.100573>
- Schmidt E, Putura PM, Fijten R (2025) The epistemic cost of opacity: how the use of artificial intelligence undermines the knowledge of medical doctors in high-stakes contexts. *Philos Technol* 38:Article 5. <https://doi.org/10.1007/s13347-024-00834-9>
- Sedlakova J, Lucivero F, Pavarini G, Kerasidou A (2025) Human-like epistemic trust? A conceptual and normative analysis of conversational AI in mental healthcare. *Am J Bioethics*. <https://doi.org/10.1080/15265161.2025.2526734>. (advance online publication)
- Smith SJ (2026) Generative artificial intelligence, formative epistemic injustice, and educators’ solutions. *Stud Philos Educ*. <https://doi.org/10.1007/s11217-026-10100-4>. (advance online publication)
- Sparrow B, Liu J, Wegner DM (2011) Google effects on memory: cognitive consequences of having information at our fingertips. *Science* 333(6043):776–778. <https://doi.org/10.1126/science.1207745>
- Storm BC, Stone SM, Benjamin AS (2017) Using the Internet to access information inflates future use of the Internet to access other information. *Memory (Hove)* 25(6):717–723. <https://doi.org/10.1080/09658211.2016.1210171>
- Tai J, Ajjaw R, Boud D, Dawson P, Panadero E (2018) Developing evaluative judgement: enabling students to make decisions about the quality of work. *High Educ* 76:467–481. <https://doi.org/10.1007/s10734-017-0220-3>
- Wineburg S, McGrew S (2019) Lateral reading and the nature of expertise: reading less and learning more when evaluating digital information. *Teach Coll Rec* 121(11):1–40. <https://doi.org/10.1177/016146811912101102>
- Wu J-Y, Lee Y-H, Chai CS, Tsai C-C (2025) Strengthening human epistemic agency in the symbiotic learning partnership with generative artificial intelligence. *Educ Res* 54(6):358–368. <https://doi.org/10.3102/0013189X251333628>
- Yan L, Pammer-Schindler V, Mills C, Nguyen A, Gašević D (2025) Beyond efficiency: empirical insights on generative AI’s impact on cognition, metacognition and epistemic agency in learning. *Br J Educ Technol* 56(5):1675–1685. <https://doi.org/10.1111/bjet.70000>
- Zawacki-Richter O, Marín VI, Bond M, Gouverneur F (2019) Systematic review of research on artificial intelligence applications in higher education: where are the educators? *Int J Educ Technol High Educ* 16:Article 39. <https://doi.org/10.1186/s41239-019-0171-0>
- Zhai C, Wibowo S, Li LD (2024) The effects of over-reliance on AI dialogue systems on students’ cognitive abilities: a systematic review. *Smart Learn Environ* 11:Article 28. <https://doi.org/10.1186/s40561-024-00316-7>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.